

Parth Pankaj Singh

New Delhi | singhparth427@gmail.com | 7318510890 | LinkedIn | Github

Education

Bennett University, B.Tech in Computer Science

2023 – 2027

- GPA: 9.13/10 (Expected graduation: Spring 2027)

Technologies

Languages: HTML, CSS, JavaScript, TypeScript, Python, C/C++, SQL

AI/ML/DL: PyTorch, Transformers, PEFT, Bitsandbytes

Python Libraries: FastAPI, Flask, OpenCV, BeautifulSoup, Pandas, LangGraph, LangChain, LlamaIndex

MLOps & Experiment Tracking: MLflow, DVC, Kubeflow, Airflow

Cloud: Azure, AWS, AWS SageMaker, Docker, Kubernetes (AKS)

Databases: MongoDB, PostgreSQL, Firebase, MySQL, Supabase, Pinecone, FAISS, ChromaDB

Web Frameworks: React.js, Next.js, Express, Node.js, Tailwind

Experience

Freelance LLM Engineer | Remote Contract

July 2025 – Sept 2025

- Fine-tuned large language models with LoRA and QLoRA for legal-domain text understanding, improving domain adaptation accuracy by 35%.
- Assessed models on case law retrieval and legal summarization, increasing domain-specific consistency by 25%.
- Engineered modular PEFT and RAG-based pipelines with Transformers, improving GPU efficiency by 10%.

Projects

NadiNetra – Multisatellite-Based Water Quality Monitoring System [GitHub](#)

Google Earth Engine, Python, PyTorch, FastAPI, MLflow, DVC, Docker, Pandas, Scikit-learn

- Orchestrated ingestion and processing of satellite imagery from Google Earth Engine to estimate key water quality parameters in near real time, reaching 99.1% accuracy across regional lakes.
- Devised a Transformer-based sequence model to infer turbidity, chlorophyll, and suspended solids from temporal satellite data.
- Operationalized an end-to-end MLOps and CI/CD workflow, automating model tracking, versioning, and deployment for scalable monitoring.

GitInspect – LLM-Powered Repository Intelligence [GitHub](#) [Live](#)

Next.js, Node.js, PostgreSQL, LangChain, Octokit

- Constructed data pipelines that ingest GitHub repositories via Octokit for commit-level and metadata analysis.
- Deployed LLM-driven commit summarization and breaking-change detection, trimming debugging effort by 35%.
- Tuned parallel LLM execution and RAG orchestration, raising throughput and reducing latency by 38%.

HealthSphere – AI-Powered Healthcare Platform [GitHub](#) [Live](#)

React.js, Node.js, MongoDB, Express.js, VideoSDK, OpenAI API, JWT

- Architected a MongoDB-backed healthcare platform supporting 500+ users with role-based access control and secure video consultations.
- Embedded OpenAI-powered report generation into clinician workflows, streamlining documentation and dashboard interactions.
- Implemented RESTful APIs and JWT authentication, automating onboarding and appointment scheduling and cutting manual coordination by 50%.

Extra Curricular / Awards

Hackathons

- Achieved top 5 position in Microsoft Engage Hackathon.
- Participated in more than 10 hackathons over 2 years with 90% finalist selection rate.

Mobilon Club Tech Lead

- Organized Hacktoberfest events, including hands-on Git and GitHub workshops promoting open-source contributions.
- Facilitated industry collaborations, guest lectures, and networking opportunities for club members.